

Sparing of the Lower Gastrointestinal Tract Microbiota Is Associated with Reduced Acute Graft-Versus-Host Disease

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Introduction. Clinical graft-versus-host disease (GVHD) phenotypes can selectively involve the upper or lower tract and extraintestinal tissues and organs. High gut microbiota diversity and the abundance of certain bacterial commensals within the Clostridiales order are associated with decreased risk of GVHD-related mortality after allogeneic stem cell transplantation (allo-HCT). The microbiota-derived short-chain fatty acid butyrate has been found beneficial in reducing murine GI aGVHD. Conversely, microbiota disruption is associated with expansions of potentially pathogenic bacteria, including facultative anaerobes, and can worsen allo-HCT outcomes. Since the vast majority of the intestinal microbiota biomass resides within the colon, we hypothesized that certain features of the colonic microbiota confer local protection to the lower GI tract.

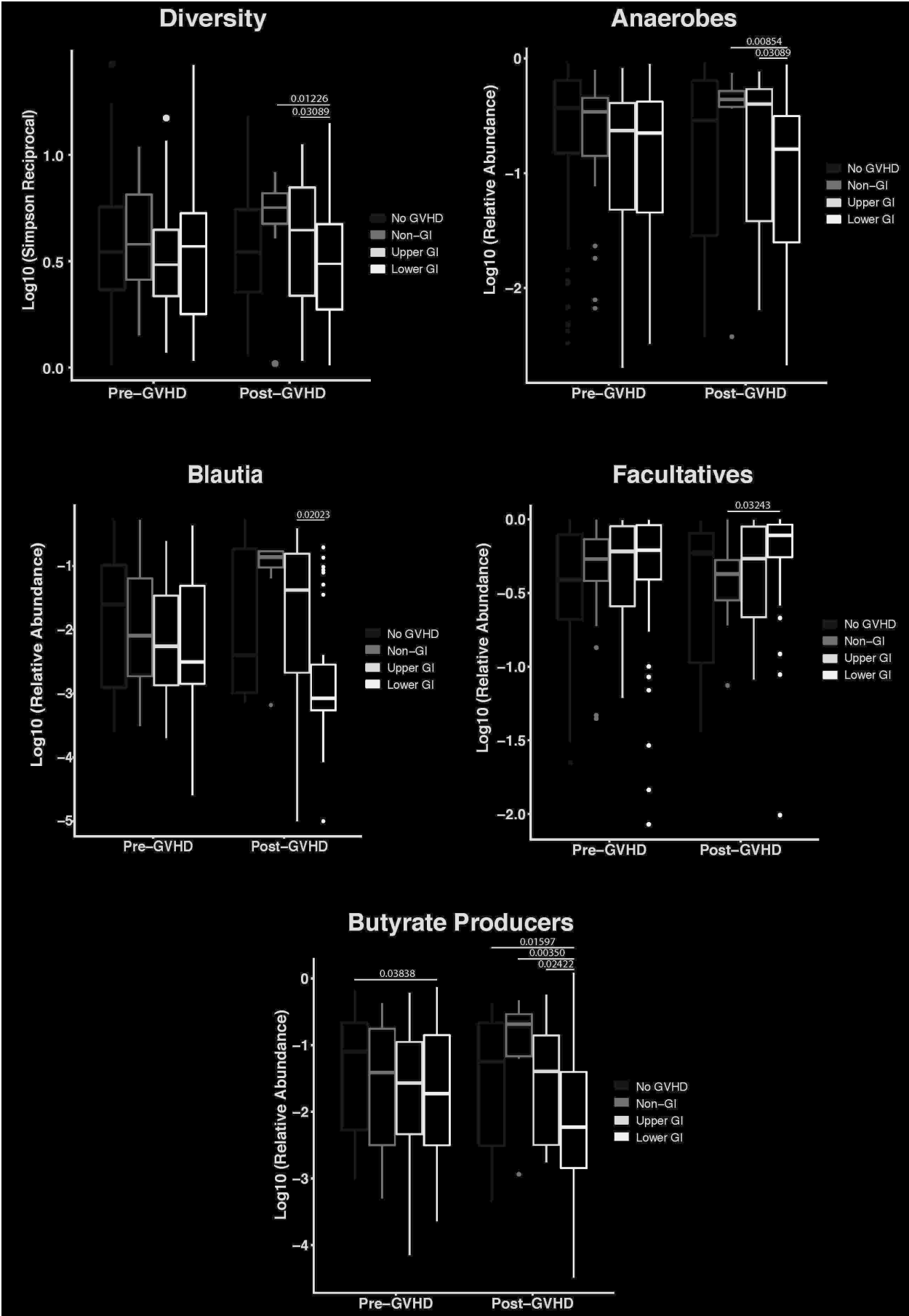
Methods. We evaluated 216 patients (median age 55 years) who underwent unmodified allo-HCT and had stool samples collected under our institutional fecal microbiome biobank between 01/2011 and 02/2017. Patients were classified in three groups according to the organ involved in aGVHD: Upper GI only (UGI, n = 56), lower GI with or without upper GI involvement (LGI, n = 61), no GI tract involvement (non-GI, n = 29). A fourth group of 70 control patients with no GVHD whatsoever were selected on the basis of the study inclusion and exclusion criteria and having evaluable samples in our biobank in a

temporal distribution that matched onset of GVHD. Microbial diversity was computed using Simpson's reciprocal index.

Results. A total of 902 stool specimens were analyzed with an average of 4.2 samples/patient. Samples were grouped into pre-onset (day -20 to day 0) and post-onset (day 1 to day 20) aGVHD. Trends in intestinal microbiota features relative to the day of aGVHD onset showed distinct dynamics among the groups. Prior to aGVHD onset, all groups showed similar microbial diversity, and similar abundance of anaerobic commensals, including members of genus *Blautia* (**Fig 1A-C**). After aGVHD onset, microbial diversity and commensal anaerobe abundance in LGI cases were significantly lower than non-GI ($p = 0.01$ and 0.008 , respectively) and UGI ($p = 0.03$ and 0.03 , respectively) groups (**Fig. 1A, 1B**). The LGI group also had significantly lower abundance of the genus *Blautia* ($p = 0.02$, **Fig 1C**). Conversely, the LGI group had increased prevalence of facultative anaerobes (LGI vs. non-GI, $p = 0.03$, **Fig. 1D**). Prior to aGVHD onset, the LGI cases had lower abundance of predicted butyrate-producing bacteria (LGI vs. no-GVHD, $p = 0.03$), which was maintained in all three GVHD groups post-onset aGVHD (**Fig 1E**).

Conclusions. Patients without GVHD, non-GI and UGI aGVHD phenotypes had higher microbial diversity, abundance of anaerobe commensals including genus *Blautia*, and predicted butyrate-producing bacteria after aGVHD onset than patients with lower GI tract involvement by aGVHD. In contrast, the LGI cases had features consistent with microbiota dysbiosis including prevalence of facultative anaerobes, and lower abundance of butyrate-producing bacteria before and after aGVHD. These findings are consistent with a model in which a diverse microbiota abundant in potential beneficial commensals exerts a local protective effect in the lower GI tract and have potential practical implications in future prophylactic and therapeutic interventions in aGVHD.

Figure 1



Disclosures

Gomes: *Seres Therapeutics*: Other: Part of Salary. **Slingerland:** *Seres Therapeutics*: Other: Salary supported by Seres funding. **Pamer:** *Bristol Myers Squibb*: Honoraria; *Celgene*: Honoraria; *Seres Therapeutics*: Honoraria, Patents & Royalties; *MedImmune*: Honoraria; *Novartis*: Honoraria; *Ferring Pharmaceuticals*: Honoraria. **Peled:** *Seres therapeutics*: Consultancy, Honoraria. **van den Brink:** *Acute Leukemia Forum (ALF)*: Consultancy, Honoraria; *Merck & Co, Inc.*: Consultancy, Honoraria; *Magenta and DKMS Medical Council*: Membership on an entity's Board of Directors or advisory committees; *Juno Therapeutics*: Other: Licensing; *Seres Therapeutics*: Consultancy, Honoraria, Membership on an entity's Board of Directors or advisory committees, Research Funding; *Amgen*: Consultancy, Honoraria; *Jazz Pharmaceuticals*: Consultancy, Honoraria; *Therakos*: Consultancy, Honoraria; *Flagship Ventures*: Consultancy, Honoraria; *Novartis*: Consultancy, Honoraria; *Evelo*: Consultancy, Honoraria.

Author notes

*Asterisk with author names denotes non-ASH members.